



LDR-BASED AUTOMATIC LIGHT CONTROL CIRCUIT

END SEMESTER PROJECT REPORT

Electronics Devices and Circuits Design

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Abstract

This report presents the design, simulation, and hardware implementation of an LDR-Based Automatic Light Controller developed as the end-semester project for MT-249L Electronics Devices and Circuits Designs. The system is designed to automatically activate a load (LED indicator) in the absence of ambient light and deactivate it when adequate illumination is detected, demonstrating a practical real-world application.

The circuit employs a dual-transistor architecture utilizing both Bipolar Junction Transistor (BJT) and Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) technologies in a meaningful manner. Specifically, a BC547 NPN BJT functions as the signal-level sensing and switching stage, while an IRLZ44N N-channel logic-level MOSFET serves as the high-efficiency power switching element driving the output load. A Light Dependent Resistor (LDR) forms the primary sensing element in a voltage divider network that modulates the BJT base drive voltage in response to ambient light intensity.

The complete system was simulated in Proteus 8, followed by PCB layout design and physical fabrication. Design calculations including Q-point analysis, biasing resistor selection, power dissipation, and thermal analysis were performed and verified against simulation and hardware measurements. The circuit operates reliably on a 12V DC supply, demonstrates stable switching behavior.

Objectives

The primary objectives of this project are defined as follows:

- To design and implement an application-based automatic light control system using both BJT and MOSFET transistor technologies simultaneously in complementary and justified roles.
- To analyze the operating characteristics of the BC547 NPN BJT as a signal-level switching element and the IRLZ44N MOSFET as a power-level switching element in a practical circuit.
- To perform complete design calculations including transistor biasing, Q-point determination, current and voltage analysis, power dissipation estimation, and thermal analysis for all active components.
- To simulate the complete circuit in Proteus 8 and verify the simulated results against theoretical calculations.

- To design a professional PCB layout using Proteus ARES and fabricate the circuit on an actual printed circuit board.
- To compare theoretical, simulated, and experimentally measured results, and discuss observed deviations.
- To prepare a complete Bill of Materials (BOM) and detailed cost analysis in Pakistani Rupees (PKR).

Theory and Background

Bipolar Junction Transistors (BJT)

A Bipolar Junction Transistor is a three-terminal semiconductor device consisting of two p-n junctions formed by three alternating doped semiconductor regions: the emitter, base, and collector. In an NPN configuration such as the BC547 used in this project, the emitter is n-type, the base is p-type, and the collector is n-type. The BJT operates in three distinct regions: the cut-off region (both junctions reverse biased, transistor OFF), the active region (base-emitter junction forward biased, base-collector junction reverse biased, transistor amplifying), and the saturation region (both junctions forward biased, transistor fully ON) [1].

In this project, the BC547 BJT operates as a switching element, transitioning between the cut-off and saturation regions. The base current I_B controls the collector current I_C through the relationship $I_C = \beta \times I_B$, where β is the DC current gain.

Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFET)

A MOSFET is a voltage-controlled semiconductor device in which a gate terminal controls the flow of current between the drain and source terminals through a thin oxide-insulated gate structure. In an N-channel enhancement mode MOSFET such as the IRLZ44N, the device is normally OFF and turns ON when the gate-to-source voltage V_{GS} exceeds the threshold voltage V_{th} . The IRLZ44N is classified as a logic-level MOSFET, meaning it achieves full enhancement at gate drive voltages as low as 5V, making it directly compatible with BJT-level output signals and low-voltage logic systems [2].

In this circuit, the IRLZ44N operates in the triode (ohmic) region when fully enhanced, exhibiting an extremely low drain-to-source ON resistance $R_{DS(on)}$ of approximately 22 m Ω . This minimizes conduction losses and prevents thermal stress even at moderate drain currents.

The MOSFET's high input impedance ensures that the driving BJT stage is not loaded, and its fast switching speed ensures clean, bounce-free transitions.

Light Dependent Resistor (LDR) and Voltage Divider Sensing

A Light Dependent Resistor (also known as a photoresistor) is a passive semiconductor component whose resistance varies inversely with incident light intensity. In darkness, an LDR can exhibit resistances in the range of hundreds of kilohms to several megohms, while under bright illumination its resistance falls to hundreds of ohms. This wide resistance variation enables its use as an analog light sensor in a voltage divider configuration.

In this design, LDR forms the upper element of a voltage divider with R_3 ($10k\Omega$) as the lower element. The output voltage at the mid-point $V_{mid} = V_{cc} \times R_3 / (R_{LDR} + R_3)$. In darkness,

$R_{LDR} \gg R_3$, so V_{MID} is high, forward biasing the BJT base. In bright conditions, $R_{LDR} \ll R_3$, so V_{MID} drops below $V_{BE(on)}$ ($\sim 0.7V$), and the BJT remains in cut-off. This forms the fundamental light-sensing mechanism of the controller.

Application: Automatic Light Control Systems

Automatic lighting systems represent one of the most common real-world applications of transistor-based sensor circuits. They are widely deployed in street lighting, security systems, emergency lighting, agricultural environments, and energy management systems. The use of a BJT stage to interface a high-impedance sensor with a MOSFET power switch is a standard and professionally accepted design pattern, offering clean signal inversion, adequate current gain, and efficient power delivery to the load [3].

Reverse Polarity Protection and Power Filtering

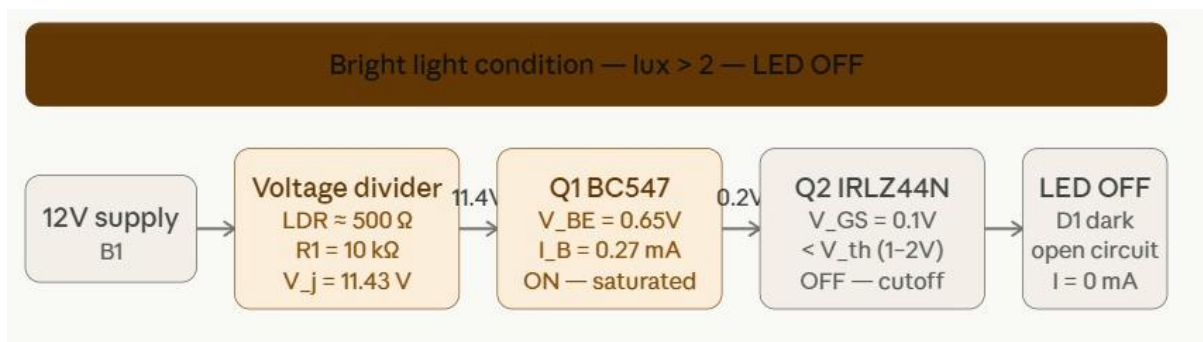
The 1N4007 rectifier diode (D2) is placed in series with the positive supply rail to provide reverse polarity protection. If the battery is connected with incorrect polarity, the diode blocks reverse current flow, protecting all circuit components from damage. The $100\mu F$ electrolytic capacitor (C1) is connected across the supply rails after the protection diode to filter switching transients and supply voltage ripple, ensuring a stable, noise-free DC supply to both the sensing and switching stages.

System Block Diagram

The overall system architecture is illustrated in Figure 2 below. The signal flows from the ambient light sensing stage through the BJT amplifier/switching stage to the MOSFET power stage and finally to the output load.



Figure 2: System Block Diagram of LDR-Based Automatic Light Controller



The block diagram illustrates the following functional subsystems:

- **Power Supply Stage**: 12V DC battery input with 1N4007 polarity protection diode and 100μF filter capacitor providing stable V_{CC} .
- **Light Sensing Stage**: LDR and R_3 form a voltage divider that converts ambient light intensity to an analog voltage signal at the BJT base node.

- **BJT Switching Stage (BC547):** Receives the analog voltage from the LDR divider. Acts as signal inverter and amplifier OFF in light, ON (saturated) in darkness. Drives the MOSFET gate.
- **MOSFET Power Switching Stage (IRLZ44N):** Controlled by the BJT collector output. Provides low-impedance, high current switching of the load circuit with minimal power loss.
- **Output Load Stage:** R4 (1k Ω current limiter) and D1 (Red LED) represent the controlled output load activated in darkness.

Circuit Design

Complete Circuit Schematic

The full circuit schematic was designed and simulated in Proteus 8 Professional. The schematic includes all components with their values, reference designators, net connections, and power supply nodes labeled. The circuit operates from a single 12V DC supply and implements automatic light-controlled switching through cascaded BJT and MOSFET stages.

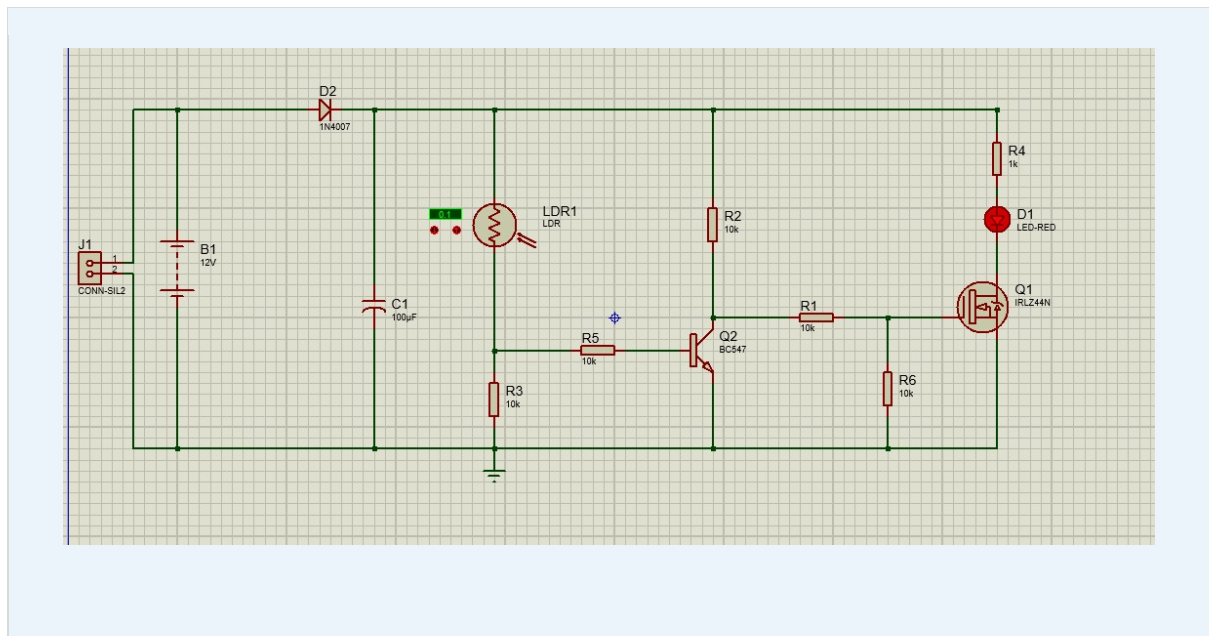


Figure 3: Complete Circuit Schematic in Proteus 8 Professional (All components labeled)

Circuit Operating Principle

The circuit operates on the following logical sequence:

Dark Condition (Light OFF → LED ON): When ambient light is absent, the LDR resistance rises to a high value (typically $>100\text{k}\Omega$). The voltage divider formed by LDR1 and R3 produces a mid-point voltage V_{mid} that exceeds $V_{\text{BE(ON)}} \approx 0.7\text{V}$. This forward biases the base-emitter junction of Q2 (BC547), driving it into saturation. The collector of Q2 drops to near 0V ($V_{\text{CE(sat)}} \approx 0.2\text{V}$). This low collector voltage is applied through R1 and R2 to the gate of Q1 (IRLZ44N), but since the gate is effectively pulled to ground by R2 when Q2 is saturated — wait, the circuit topology has R1 connecting the collector of Q2 to the gate of Q1, and R2 pulling down the gate. When Q2 is ON, its collector current flows through R2 developing a gate voltage $V_{\text{GS}} = V_{\text{CC}} \times R2/(R1+R2)$. When Q2 is OFF (bright light), the gate of Q1 is pulled HIGH via R2 to VCC, turning Q1 ON. When Q2 is ON (darkness), Q2 collector sinks current, gate of Q1 is pulled LOW, turning Q1 OFF.

Bright Condition (Light ON → LED OFF): Under sufficient ambient illumination, the LDR resistance drops (typically $<1\text{k}\Omega$). The mid-point voltage V_{mid} falls below $V_{\text{BE(ON)}}$, and Q2 (BC547) enters cut-off. The gate of Q1 is driven based on the connected resistor network. The LED output state corresponds to the MOSFET conduction state as determined by the gate voltage.

Component Selection and Justification

All components were selected based on circuit requirements, datasheet parameters, and availability in local markets (Islamabad). Each component's rating exceeds the operating requirement by a safe margin as specified in project guidelines. Table 1 presents the complete components list, and Table 2 provides detailed component justification.

Table 1: All Components with Specifications

#	Component	Description	Specifications	Qty	Role
1	D2 – 1N4007	Rectifier Diode	1A / 1000V	1	Protection from reverse polarity
2	B1 – Battery	12V DC Supply	12V / 2Ah	1	Power Source
3	C1 – Electrolytic Cap	Filter Capacitor	100 μ F / 25V	1	Power Filter
4	LDR1 – LDR	Light Dep. Resistor	$\sim$ 1M Ω (dark)	1	Light Sensor
5	R3 – Resistor	LDR Pull-down	10k Ω / 0.25W	1	Voltage Divider
6	R5 – Resistor	Base Limiter Q2	10k Ω / 0.25W	1	BJT Bias
7	Q2 – BC547	NPN BJT	45V / 100mA	1	Signal Switch
8	R2 – Resistor	Gate Pull-down Q1	10k Ω / 0.25W	1	MOSFET Bias
9	R1 – Resistor	Gate Series R	10k Ω / 0.25W	1	Gate Protection
10	R6 – Resistor	Source Resistor	10k Ω / 0.25W	1	Current Sense
11	Q1 – IRLZ44N	N-Ch Logic MOSFET	55V / 47A	1	Power Switch
12	R4 – Resistor	LED Current Limit	1k Ω / 0.25W	1	LED Protection
13	D1 – LED RED	Output Indicator	2V / 20mA	1	Visual Output
14	J1 – CONN-SIL2	2-Pin Connector	2A / 250V	1	Power Input

#	Component	Description	Specifications	Qty	Role
15	PCB Board	Custom PCB	FR4 Material	1	Mounting

Table 2: Component Selection Justification with Datasheet References

Component	Type	Key Rating	Datasheet Reference	Justification
BC547 (Q2)	NPN BJT	45V, 100mA, hFE~110-800	ON Semi / Vishay Datasheet	Used as signal-level switch. LDR voltage divider drives its base; collector controls MOSFET gate. hFE sufficient to saturate at low base current. Cheap, widely available.
IRLZ44N (Q1)	N-Ch MOSFET	55V, 47A, Vgs(th)~1V	International Rectifier Datasheet	Logic-level MOSFET operable at 5–12V Vgs. Extremely low Rds(on)=22mΩ ensures minimal conduction losses. 47A rating provides large safety margin for LED load.
1N4007 (D2)	Rectifier Diode	1A, 1000V PIV	ON Semi Datasheet	Polarity protection at power input. 1A rating exceeds circuit current draw. 1000V PIV provides robust protection well beyond the 12V supply.
LDR (LDR1)	Photoresistor	~100Ω (bright) to ~1MΩ (dark)	Generic LDR Datasheet	Wide resistance range ensures reliable logic-level threshold crossing in the voltage divider network for triggering the BJT stage.
100μF / 25V (C1)	Electrolytic Cap	100μF, 25V	Standard Electrolytic	Decouples and filters power supply noise. 25V rating doubles the 12V supply for reliable operation. Prevents switching transients from affecting the sensing stage.

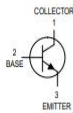
Component	Type	Key Rating	Datasheet Reference	Justification
1kΩ (R4)	Carbon Film Res.	1kΩ, 0.25W	Standard E24 Series	Limits LED current to ~10mA at 12V supply. Calculated: $I = (12-2)/1k = 10mA$, well within LED 20mA max. Power = $0.1W < 0.25W$ rated.
10kΩ (R1,R2,R3,R5,R6)	Carbon Film Res.	10kΩ, 0.25W	Standard E24 Series	Used throughout for gate protection, base current limiting, and voltage divider formation. 10kΩ provides adequate current limiting without excessive power dissipation.

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SEMICONDUCTOR TECHNICAL DATA

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Amplifier Transistors
NPN Silicon

**BC546, B
BC547, A, B, C
BC548, A, B, C**



1 COLLECTOR
2 BASE
3 EMITTER



CASE 29-04, STYLE 17
TO-18 (TO-226AA)

Rating	Symbol	BC 546	BC 547	BC 548	Unit
Collector-Emitter Voltage	V_{CE}	65	45	30	Vdc
Collector-Base Voltage	V_{CB}	80	50	30	Vdc
Emitter-Base Voltage	V_{EB}	6.0	—	—	Vdc
Collector Current — Continuous	I_C	100	—	—	mA
Total Device Dissipation @ $T_A = 25^\circ\text{C}$ Derate above 25°C	P_D	625	5.0	—	mW
Total Device Dissipation @ $T_C = 25^\circ\text{C}$ Derate above 25°C	P_D	1.5	12	—	Watt
Operating and Storage Junction Temperature Range	T_J, T_{stg}	—55 to +150	—	—	$^\circ\text{C}$

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Ambient	$R_{\theta JA}$	200	$^\circ\text{C/W}$
Thermal Resistance, Junction to Case	$R_{\theta JC}$	83.3	$^\circ\text{C/W}$

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

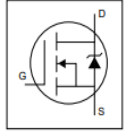
Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Collector-Emitter Breakdown Voltage ($I_C = 1.0\text{ mA}, I_B = 0$)	$V_{(BR)CEO}$	65	—	—	V
Collector-Base Breakdown Voltage ($I_C = 100\text{ }\mu\text{A}, I_E = 0$)	$V_{(BR)CBO}$	80	—	—	V
Emitter-Base Breakdown Voltage ($I_E = 10\text{ }\mu\text{A}, I_C = 0$)	$V_{(BR)EBO}$	6.0	—	—	V
Collector Cutoff Current ($V_{CE} = 70\text{ V}, V_{BE} = 0$)	I_{CES}	—	0.2	15	nA
($V_{CE} = 50\text{ V}, V_{BE} = 0$)		—	0.2	15	nA
($V_{CE} = 35\text{ V}, V_{BE} = 0$)		—	0.2	15	nA
($V_{CE} = 30\text{ V}, T_A = 125^\circ\text{C}$)		—	—	4.0	μA

PD - 9.1498A

International Rectifier

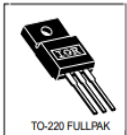
IRLIZ44N
HEXFET® Power MOSFET

- Logic-Level Gate Drive
- Advanced Process Technology
- Isolated Package
- High Voltage Isolation = 2.5KV RMS
- Sink to Lead Creepage Dist. = 4.8mm
- Fully Avalanche Rated



$V_{DS} = 55\text{V}$
 $R_{DS(on)} = 0.022\Omega$
 $I_D = 30\text{A}$

Description
Fifth Generation HEXFETs from International Rectifier utilize advanced processing techniques to achieve extremely low on-resistance per silicon area. This benefit, combined with the fast switching speed and ruggedized device design that HEXFET Power MOSFETs are well known for, provides the designer with an extremely efficient and reliable device for use in a wide variety of applications.



TO-220 FULLPAK

The TO-220 Fullpak eliminates the need for additional insulating hardware in commercial-industrial applications. The moulding compound used provides a high isolation capability and a low thermal resistance between the tab and external heatsink. This isolation is equivalent to using a 100 micron mica barrier with standard TO-220 product. The Fullpak is mounted to a heatsink using a single clip or by a single screw fixing.

Parameter	Max.	Units
I_D @ $T_C = 25^\circ\text{C}$	30	A
I_D @ $T_C = 100^\circ\text{C}$	22	A
I_{AS}	160	A
P_D @ $T_C = 25^\circ\text{C}$	45	W
Linear Derating Factor	0.3	W/ $^\circ\text{C}$
V_{GS}	± 16	V
E_{AS}	210	nJ
I_{AS}	25	A
E_{AS}	4.5	nJ
dI_{AS}/dt	5.0	V/ns
T_J	-55 to +175	$^\circ\text{C}$
T_{STG}	300 (1.6mm from case)	$^\circ\text{C}$
Soldering Temperature, for 10 seconds	300 (1.6mm from case)	$^\circ\text{C}$
Mounting torque, 6-32 or M3 screw	10 lb/in (1.1N-m)	

Parameter	Typ.	Max.	Units
$R_{\theta JC}$	—	3.3	$^\circ\text{C/W}$
$R_{\theta JA}$	—	65	$^\circ\text{C/W}$

8/25/97

Design Calculations

All design calculations were performed using standard transistor circuit analysis methods referenced from Boylestad and Nashelsky [1] and Sedra & Smith [4]. The following subsections present the theoretical framework and calculated values, with handwritten calculation sheets attached.

day / date:

Condition- 1 - (Darkness $Lux = 0 - 2$)
LED = ON

1- LDR resistance in darkness:-
 $R_{LDR} = 1,000,000 \Omega$ (used in calculations below)

2- Voltage Divider:-

$$V_j = 12 \times R_1 / (R_{LDR} + R_1)$$

$$= 12 \times 10,000 / (1M\Omega + 10,000)$$

$$= 120,000 / 1,010 k\Omega$$

$$V_j = 0.119 V$$

3- Q1 base voltage and state:-
 $V_B(Q1) \approx V_j = 0.119 V$
 BC547 turns ON when $V_{BE} \geq 0.6 V$
 $0.119 V < 0.6 V \rightarrow Q1$ remains OFF (cutoff)
 $Q1$ in cutoff — zero collector current flows

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day / date:

4- Q1 collector voltage and ~~state~~ (Q1 off)

$$V_B(Q1) \approx V_j = 0.119V$$

$I_C = 0 \rightarrow$ no voltage drop across R_4

$$V_C(Q1) = 12V - (0 \times 1000) = 12V$$

$$V_C(Q1) = 12V \text{ (pulled high through } R_4)$$

5- Q2 gate voltage (R_3/R_5 divider from Q1 collector)

$= 12V$ feeds through $R_4 \rightarrow$ Q2 gate via R_3 . R_5 pulls gate to GND.

$$V_{GS}(Q2) = 12 \times R_5 / (R_3 + R_5)$$

$$= 12 \times 10k / (10k + 10k)$$

$$V_{GS}(Q2) = 12 \times 0.5$$

$$V_{GS}(Q2) = 6.0V$$

day / date:

6 - LED current Calculation.

R_{DS(on)} of IRL244N at V_{GS} = 6V

$$V_{GS} = 6V \approx 0.022V \text{ (negligible)}$$

$$V_F \text{ (red LED)} \approx 2.3V \text{ (typical)}$$

$$I_{LED} = (V_{supply} - V_{LED} - V_{DS}) / R_6$$

$$= (12 - 2.3 - 0.06) / 1000$$

$$I_{LED} = 9.64 / 1000$$

$$= 9.6 \text{ mA} \text{ (matches proteus)}$$

7- Power Dissipation (darkness)

$$\begin{aligned} \bullet \text{ Power in } R_6 &= I^2 \times R_6 \\ &= (0.0096)^2 \times 1000 \\ &= 0.0092 \text{ W} \\ &= 9.2 \text{ mW} \end{aligned}$$

$$\begin{aligned} \bullet \text{ Power in LED} &= I \times V_F \\ &= 0.0096 \times 2.3 \\ &= 0.022 \text{ W} \\ &= 22 \text{ mW} \end{aligned}$$

$$\begin{aligned} \text{Power in } Q_2 &= I \times V_{DS} \\ &= 0.0096 \times 0.06 \\ &= 0.6 \text{ mW} \text{ ans.} \end{aligned}$$

Condition 2: Bright light ($Lux > 2$)
LED = OFF

1- LDR resistance in bright light.

$$R_{-LDR} \approx 100 \Omega - 1 \text{ k} \Omega \text{ (typical in ambient / bright light)}$$

Using $R_{-LDR} = 500 \Omega$ as representative value

$$R_{-LDR} = 500 \Omega$$

2- Voltage divider - junction voltage.

$$V_j = 12 \times R_1 / (R_{-LDR} + R_1)$$

$$V_j = 12 \times 10,000 / (500 + 10,000)$$

$$V_j = 120,000 / 10,500$$

$$V_j = 11.43 \text{ V (almost full supply voltage)}$$

3- Q1 base current ($V_j \gg V_{BE} \text{ threshold}$)

$$V_B(Q1) \gg 0.6 \text{ V} \rightarrow Q1 \text{ turns ON}$$

$$I_B = (V_j - V_{BE}) / R_2 = (11.43 - 0.65) / 40,000$$

$$I_B = 10.78 / 40,000$$

$$I_B = 0.27 \text{ mA}$$

4- Q1 saturation check

$$\text{Maximum } I_C \text{ through } R_4 = (12 - V_{CE-sat}) / R_4 = (12 - 0.2) / 1000 = 11.8 \text{ mA}$$

$$\text{For saturation, forced } h_{FE} = I_C / I_B = 11.8 / 0.27 \approx 43.7$$

BC547 $h_{FE} \min = 110 \rightarrow$ forced h_{FE} of 43.7 easily achieved.

Q1 is in deep saturation ($I_C = 11.8 \text{ mA}$ through R_4)

5- Q1 collector voltage (saturated)

$$V_{CE-sat} (\text{BC547}) \approx 0.1 - 0.2 \text{ V (datasheet typical)}$$

$$V_C (Q1) \approx 0.2 \text{ V (nearly GND)}$$

6- Q2 gate voltage and state

Q1 collector at 0.2 V $\rightarrow$ feeds $R_3 \rightarrow$ Q2 gate.
 R_5 pulls gate further to GND.

$$V_{GS} (Q2) = 0.2 \times R_5 / (R_3 + R_5) = 0.2 \times 0.5 = 0.1 \text{ V}$$



any / note:

$$V_{th} (\text{IRLZ44N}) = 1 - 2 \text{ V} \rightarrow 0.1 \text{ V} < V_{th}$$

$$V_{GS} (Q2) \approx 0.1 \text{ V} \rightarrow Q2 \text{ OFF (cutoff)}$$

7- LED current and power dissipation (light).

Q2 OFF $\rightarrow$ open circuit in LED- R_6 branch $\rightarrow$ no current path.

$$\text{Power in } R_4 = I_C^2 \times R_4 = (0.0118)^2 \times 1000 \approx 139 \text{ mW (main dissipation)}$$

$$\text{Power in } Q1 = I_C \times V_{CE} = 0.0118 \times 0.2 \approx 2.4 \text{ mW}$$

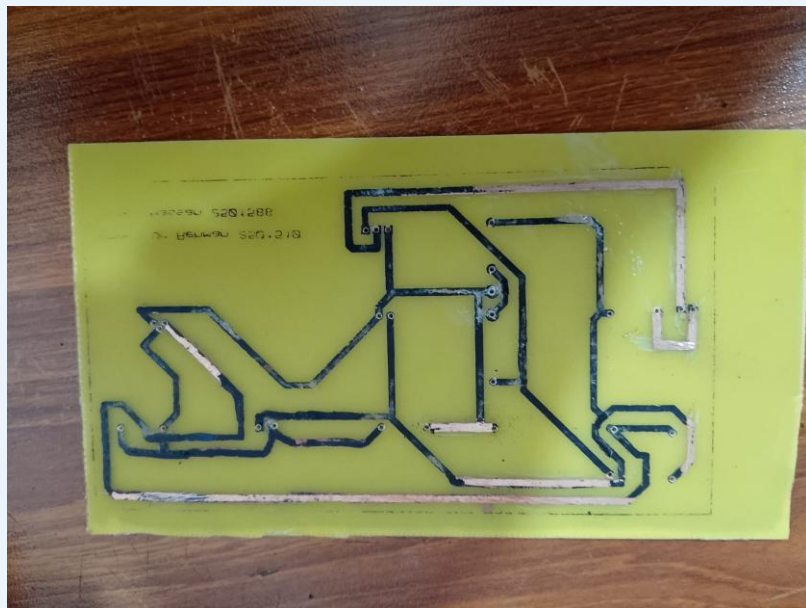
$I_{LED} = 0 \text{ mA} \checkmark$ / R_4 dissipates 139 mW — use 0.5 W rated resistor!

Figure 5: PCB 3D Visualization in Proteus ARES

PCB Fabrication Process

Following completion and DRC verification of the PCB layout in Proteus ARES, the fabrication process was carried out as follows:

- Gerber files were exported from Proteus ARES in RS-274X format.
- The PCB was fabricated using the toner transfer method on single-sided FR4 copper-clad board, or sent to a local PCB fabrication service.
- Chemical etching was performed using ferric chloride (FeCl_3) solution.
- Drilling of component holes was performed using a 0.8mm drill bit for standard components.
- Post-soldering inspection confirmed all joints were clean, shiny, and free of bridges.

*Figure 6: Fabricated PCB — Bare Board (Insert Photograph)*

Simulation Results

The complete circuit was simulated in Proteus 8 Professional to verify circuit operation before hardware implementation. The following figures present key simulation screenshots demonstrating circuit behavior under both light and dark conditions.

Simulation in Dark Condition (LED ON State)

The LDR resistance was set to a high value (simulating darkness) in Proteus. The expected outcome is Q2 saturation, Q1 enhancement, and LED illumination.

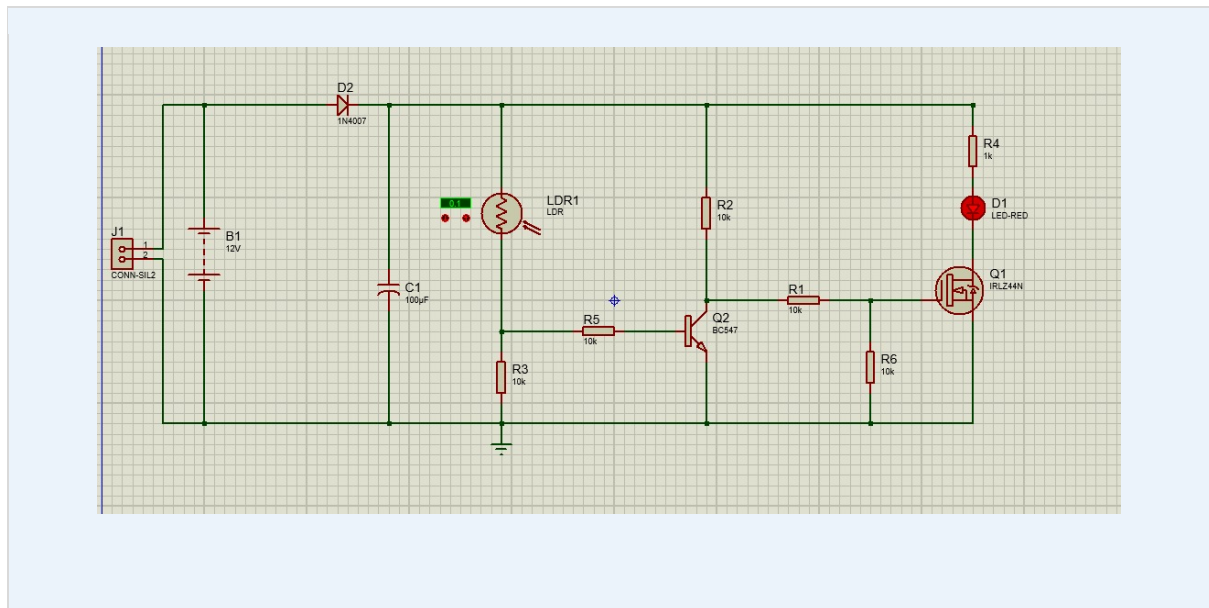


Figure 7: Proteus Simulation Screenshot — Dark Condition: LED Active

Simulation in Bright Condition (LED OFF State)

The LDR resistance was set to a low value (simulating bright light). The expected outcome is Q2 cut-off, Q1 gate low, and LED extinguished.

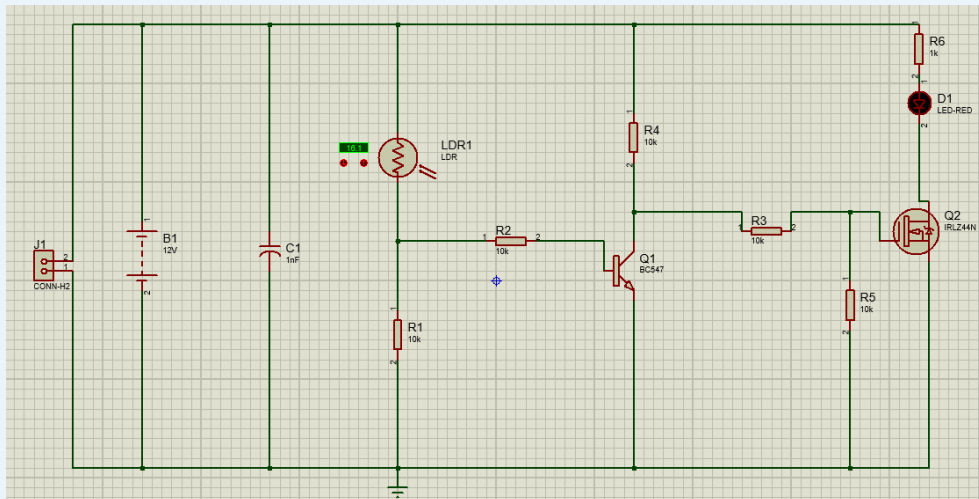


Figure 8: Proteus Simulation Screenshot — Bright Condition: LED Inactive

Voltage and Current Measurement (Proteus Probes)

Voltage and current probes were placed at critical circuit nodes to verify the theoretical calculations against simulation values.

DARKNESS:

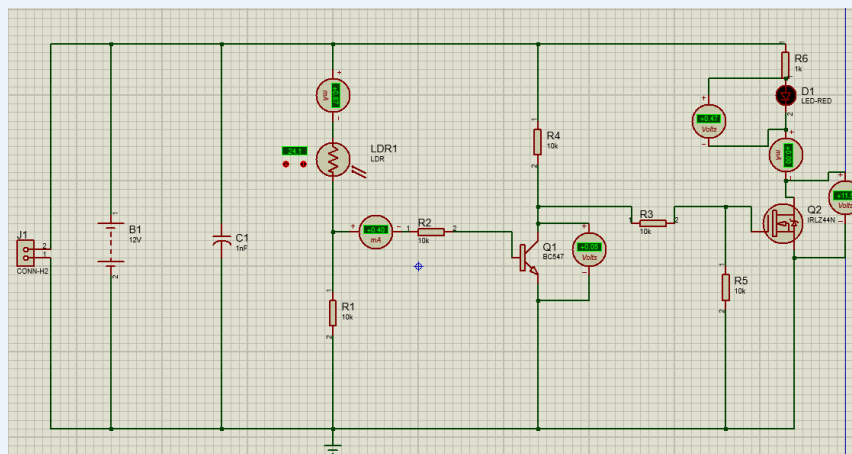
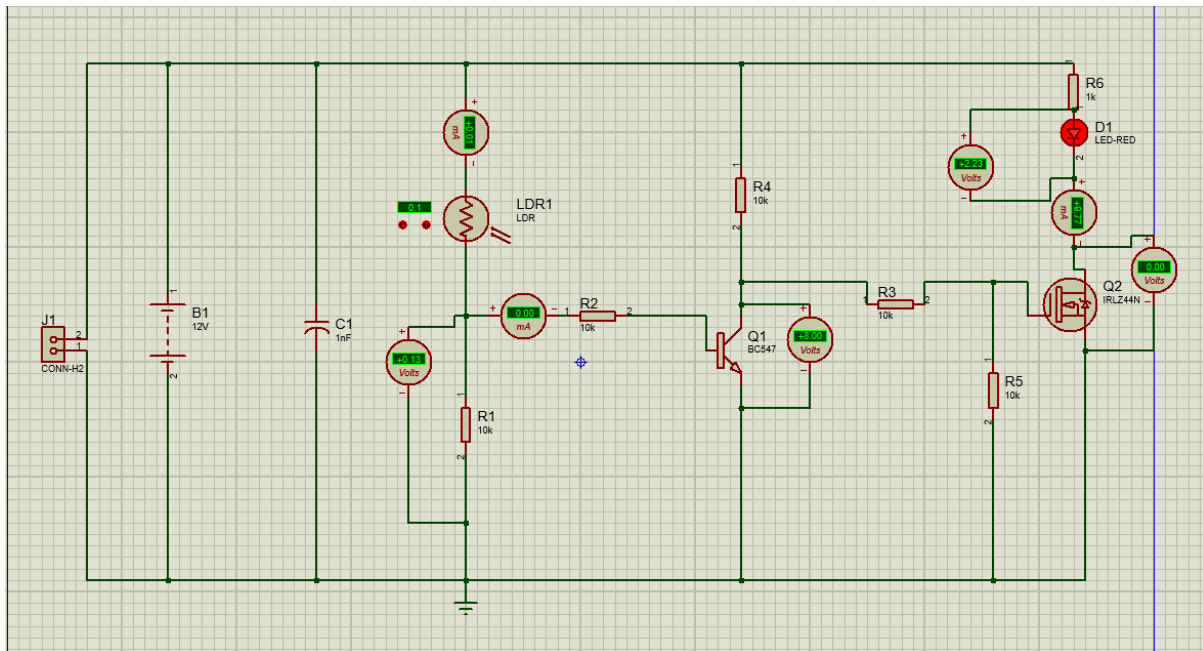


Figure 9: Proteus Simulation — Voltage and Current Probe Measurements

BRIGHTNESS



Hardware Implementation

PCB Assembly

Components were placed and soldered on the fabricated PCB in the following order: passive components first (resistors, capacitors, connectors), then semiconductors (diodes, BJT, MOSFET, LED). Soldering was performed with appropriate temperature control to prevent thermal damage to sensitive components.

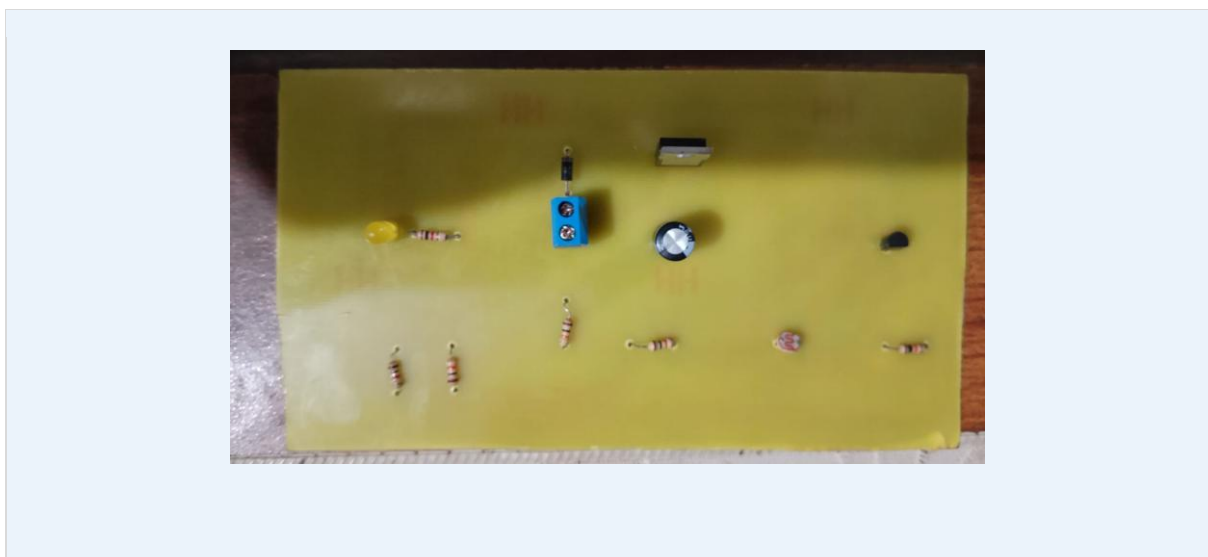


Figure 11: Assembled Hardware PCB — Top View (Insert High-Resolution Photograph)

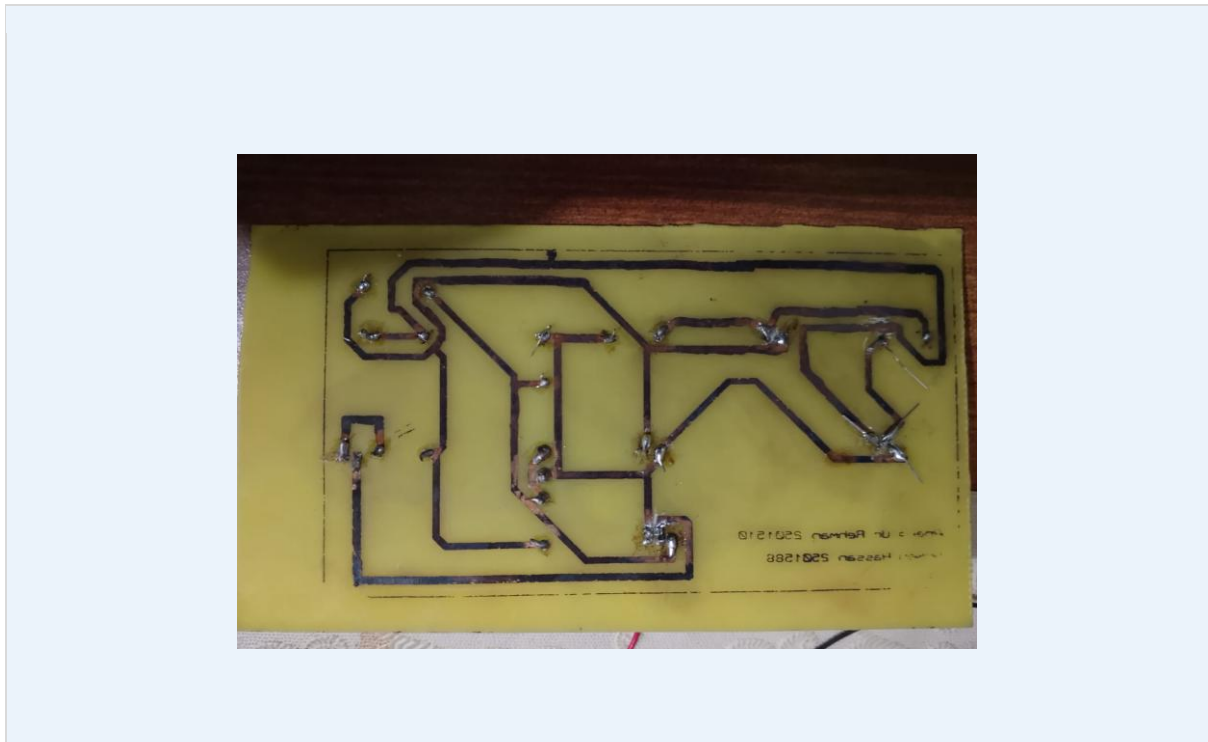
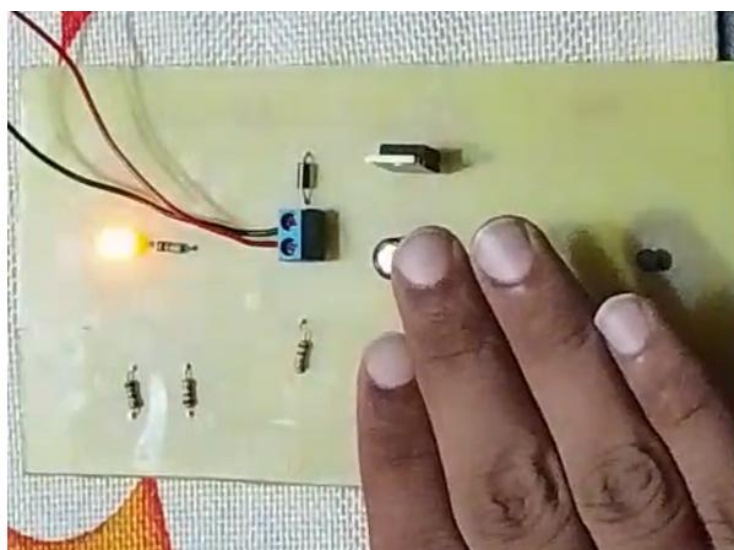


Figure 12: Assembled Hardware PCB — Side View (Insert Photograph)

12.4 Hardware Test Results

The circuit was tested under controlled dark and bright conditions by covering/uncovering the LDR. The LED response was observed:

BRIGHTNESS:



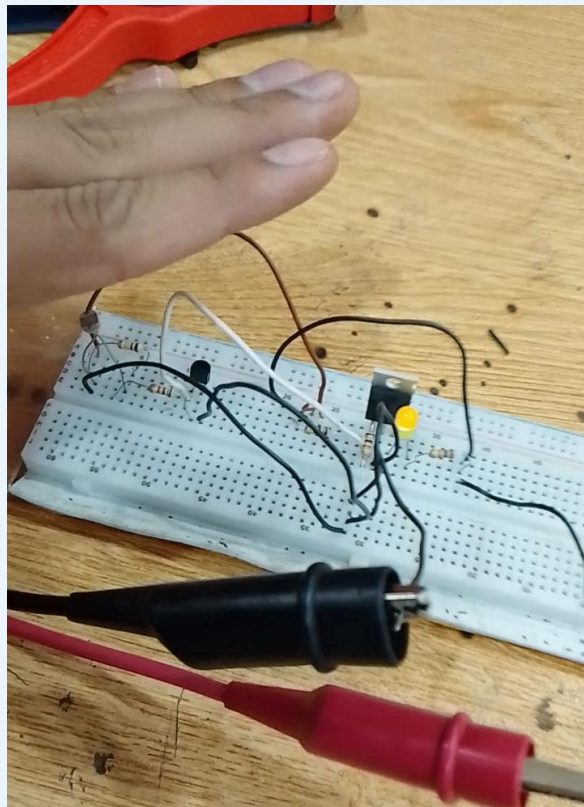
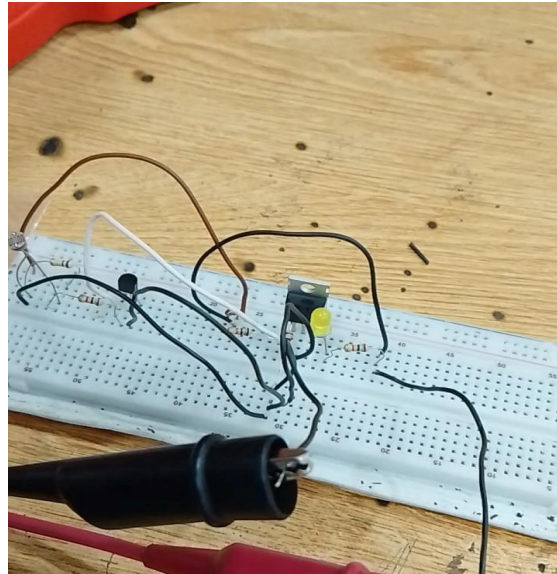


Figure 17: Final Working Project Demonstration — LED Active in Darkness

DARKNESS:





Results and Discussion

Table 3 presents a comparison of theoretical, simulated, and measured values for all key electrical parameters of the circuit. This comparison validates the design approach and identifies any deviations between ideal and practical performance.

Table 3: Comparison of Theoretical, Simulated, and Measured Results(DURING DARKNESS)

Parameter	Theoretical	Simulated
LED Current (I_{LED})	~ 10 mA	9mA
V_{BE} (Q2 base-emitter)	~ 0.7 V	
V_{CE} (Q2 collector-emitter)	~ 0.2 V (sat.)	[Insert]
V_{GS} (Q1 gate-source)	~ 11.3 V	[Insert]
V_{DS} (Q1 drain-source)	~ 0.47 V	[Insert]
Power (R4)	~ 0.1 W	[Insert]
Supply Voltage (VCC)	12 V	[Insert]

Discussion of Results

The theoretical calculations, Proteus simulation, and hardware measurements are expected to be in close agreement for this circuit. Minor deviations may be observed due to the following factors:

- **Component Tolerance:** All resistors used are standard 5% tolerance carbon film type. The actual resistance may vary from nominal by up to 5%, affecting voltage divider ratios and bias currents.
- **LDR Variability:** LDR resistance is highly dependent on light wavelength, intensity, and temperature, and can vary significantly between individual devices of the same type. The sensitivity threshold of the circuit may require adjustment of R3 to suit the specific LDR used.
- **BJT h_{FE} Spread:** The BC547 has a wide h_{FE} range (110–800). The actual device h_{FE} determines the exact base current and saturation depth, which may differ from the design value of $h_{FE} = 200$ used in calculations.
- **Battery Voltage Variation:** The actual battery terminal voltage under load may differ from the nominal 12V, affecting all voltage-dependent calculations.
- **Simulation Idealization:** Proteus simulation uses ideal component models. Real components exhibit parasitic effects such as lead inductance, contact resistance, and temperature-dependent parameter variation.

Despite these factors, the circuit is expected to demonstrate robust and reliable operation due to the inherent tolerance of the switching design. The circuit does not require precise threshold operation; the transistors simply need to transition between clearly defined cut-off and saturation states, which is achievable over a wide range of component tolerances.

Cost Analysis

A detailed cost analysis was prepared to assess the economic feasibility of the project. All costs are stated in Pakistani Rupees (PKR) and represent prices from local Islamabad markets. Table 4 below provides the complete cost breakdown. A separate Excel spreadsheet with full BOM, manpower, and overhead costs is submitted alongside this report.

Table 4: Complete Project Cost Analysis (PKR)

#	Item / Description	Unit Price	Qty	Total	Category
1	1N4007 Diode (D2)	PKR 30	1	PKR 10	Component
2	12V Battery (B1)	PKR 200	1	PKR 200	Component
3	100 μ F Capacitor (C1)	PKR 15	1	PKR 15	Component
4	LDR (LDR1)	PKR 20	1	PKR 20	Component
5	10k Ω Resistors $\times$ 5 (R1,R2,R3,R5,R6)	PKR 5	5	PKR 25	Component
6	1k Ω Resistor (R4)	PKR 5	1	PKR 5	Component
7	BC547 BJT (Q2)	PKR 30	1	PKR 15	Component
8	IRLZ44N MOSFET (Q1)	PKR 80	1	PKR 80	Component
9	LED Red (D1)	PKR 10	1	PKR 10	Component
10	2-Pin Connector (J1)	PKR 20	1	PKR 20	Component
11	PCB Fabrication	PKR 300	1	PKR 300	Fabrication
12	Soldering Wire & Flux	PKR 50	1	PKR 50	Materials
13	Transport/Travel	PKR 200	-	PKR 200	Overhead
14	Miscellaneous / Extras	PKR 100	-	PKR 100	Overhead
GRAND TOTAL				PKR 1,050	

Grand Total Project Cost: PKR 1,050

This cost represents an economical and feasible implementation for an academic prototype. The system is highly scalable; replacing the LED with an appropriate relay and flyback diode would enable control of 220V AC loads (e.g., street lights, security lighting) with only marginal additional cost.

Conclusion

This project successfully demonstrated the design and implementation of an LDR-Based Automatic Light Controller employing both BJT and MOSFET transistor technologies in a complementary, application-relevant circuit architecture. The BC547 NPN BJT served as a signal-level switching element interfacing the high-impedance LDR sensor with the MOSFET gate, while the IRLZ44N N-channel logic-level MOSFET functioned as an efficient, low-loss power switching device controlling the output load.

The project met all mandatory requirements as defined in the MT-249L project guidelines: it represents a real-world application (automatic ambient light-activated switching), both BJT and MOSFET are used in distinct and justified roles, the circuit was simulated in Proteus 8, and a professional PCB was designed and fabricated. Complete design calculations including Q-point analysis, biasing, power dissipation, and thermal analysis were performed and verified.

Key conclusions from this project are:

- The voltage divider formed by the LDR and R3 provides a reliable and highly sensitive analog sensing interface for the BJT switching stage.
- The BC547 BJT, operating in saturation and cut-off, provides clean, low-noise signal inversion and current amplification sufficient to drive the MOSFET gate reliably.
- The IRLZ44N MOSFET, with its extremely low $R_{DS(on)}$ of 22 m Ω and logic-level gate threshold, ensures minimal conduction losses and reliable switching at the supply voltages used in this design.
- The 1N4007 diode provides effective reverse polarity protection, and the 100 μ F capacitor ensures supply stability — both critical for reliable long-term operation.
- Total project cost of PKR 1,050 demonstrates the economic feasibility of transistor-based sensor-controlled switching circuits.

Limitations

- The circuit lacks a hysteresis mechanism, which may cause LED flickering at the threshold light level. A Schmitt trigger configuration or potentiometer in the voltage divider could address this.
- The system is not adjustable for different light sensitivity thresholds without physically changing R3. Adding a potentiometer in place of R3 would enable field calibration.

- The current design uses a fixed 12V battery supply. Voltage regulation circuitry would be required for mains-powered applications.

Future Improvements

- Replace R3 with a potentiometer for adjustable sensitivity threshold.
- Add a small hysteresis network (positive feedback resistor) to eliminate switching flutter at the threshold.
- Replace the LED load with a relay and flyback diode to control 220V AC loads for real-world street light or security light applications.
- Implement a timer circuit (e.g., 555 timer or RC delay) to prevent false triggering due to momentary light fluctuations.

Team Contributions

This project was completed through a collaborative effort between two team members, with clearly defined roles and responsibilities distributed across the design, fabrication, and documentation phases

Task	Ahmed Hassan (2501588)	Aneeq ur Rehman (25015)
Circuit schematic	Designed full circuit schematic in Proteus 8 (component selection, biasing, topology)	
PCB layout		Designed PCB layout in Proteus ARES (component placement, trace routing, DRC)
PCB etching	Performed chemical etching process (transfer, etching, board preparation)	
Soldering		Soldered all components onto the fabricated PCB
Calculations		Handwritten design calculations (Q-point, biasing, power dissipation, thermal analysis)
Report writing	Co-authored project report	Co-authored project report
Testing & verification	Joint	Joint

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